JPA-Part8

Till now, our *Repository interface* looks like this

```
@Repository
public interface UserDetailsRepository extends
JpaRepository<UserDetails, Long> {

}

```

And in service class, we used to invoke methods which are available in JPA framework

```
@Service
public class UserDetailsService {

    @Autowired
    UserDetailsRepository userDetailsRepository;

    public UserDetails saveUser(UserDetails user) {
        return userDetailsRepository.save(user);
    }

    public UserDetails findByID(Long primaryKey) {
        return userDetailsRepository.findById(primaryKey).get();
    }
}

```

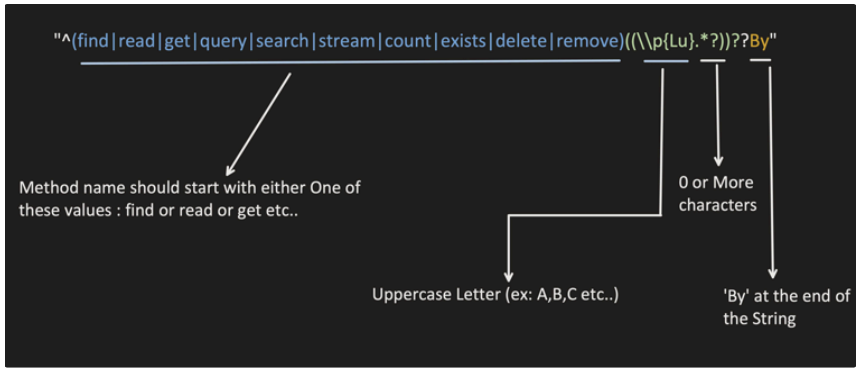
Concept && Coding

Then, we have something called: **Derived Query**

- Automatically generates queries from the methods.
- Need to follow a specific naming convention.
- Derived query used for GET/REMOVE operations but not for INSERT/UPDATE
  - Insert and Update operations is supported though "save()"

PartTree.java

```
private static final String QUERY_PATTERN = "find|read|get|query|search|stream";
private static final String COUNT_PATTERN = "count";
private static final String EXISTS_PATTERN = "exists";
private static final String DELETE_PATTERN = "delete|remove";
private static final Pattern PREFIX_TEMPLATE = Pattern.compile( //
        "^(" + QUERY_PATTERN + "|" + COUNT_PATTERN + "|" + EXISTS_PATTERN + "|" + DELETE_PATTERN + ")((\\p{Lu}.*?))??By");
```




```java
@Repository
public interface UserDetailsRepository extends
        JpaRepository<UserDetails, Long> {

    List<UserDetails> findUserDetailsByName(String userName);
}

```

```java
@Table(name = "user_details")
@Entity
public class UserDetails {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long userId;

    @Column(name = "user_name")
    private String name;

    private String phone;

    // getters and setters
}

```

Concept && Coding

Query in which it get translates too:

```
Hibernate:
    select
        ud1_0.user_id,
        ud1_0.user_name,
        ud1_0.phone
    from
        user_details ud1_0
    where
        ud1_0.user_name=?
```

<u>Different Use cases:</u>

<u>And:</u>

```java
List<UserDetails> findUserDetailsByNameAndPhone(String userName, String phone);

```

```
Hibernate:
    select
        ud1_0.user_id,
        ud1_0.user_name,
        ud1_0.phone
    from
        user_details ud1_0
    where
        ud1_0.user_name=?
        and ud1_0.phone=?
```

Or:

```java
List<UserDetails> findUserDetailsByNameAndPhoneOrUserId(String userName, String phone, Long id);

```

```
Hibernate:
    select
        ud1_0.user_id,
        ud1_0.user_name,
        ud1_0.phone
    from
        user_details ud1_0
    where
        ud1_0.user_name=?
        and ud1_0.phone=?
        or ud1_0.user_id=?
```

Concept && Coding

Comparison:

Part.java

```
BETWEEN(2, "IsBetween", "Between"),
IS_NOT_NULL(0, "IsNotNull", "NotNull"),
IS_NULL(0, "IsNull", "Null"),
LESS_THAN("IsLessThan", "LessThan"),
LESS_THAN_EQUAL("IsLessThanEqual", "LessThanEqual"),
GREATER_THAN("IsGreaterThan", "GreaterThan"),
GREATER_THAN_EQUAL("IsGreaterThanEqual", "GreaterThanEqual"),
BEFORE("IsBefore", "Before"),
AFTER("IsAfter", "After"),
NOT_LIKE("IsNotLike", "NotLike"),
LIKE("IsLike", "Like"),
STARTING_WITH("IsStartingWith", "StartingWith", "StartsWith"),
ENDING_WITH("IsEndingWith", "EndingWith", "EndsWith"),
IS_NOT_EMPTY(0, "IsNotEmpty", "NotEmpty"),
IS_EMPTY(0, "IsEmpty", "Empty"),
NOT_CONTAINING("IsNotContaining", "NotContaining", "NotContains"),
CONTAINING("IsContaining", "Containing", "Contains"),
NOT_IN("IsNotIn", "NotIn"),
IN("IsIn", "In"),
NEAR("IsNear", "Near"),
WITHIN("IsWithin", "Within"),
REGEX("MatchesRegex", "Matches", "Regex"),
EXISTS(0, "Exists"),
TRUE(0, "IsTrue", "True"),
FALSE(0, "IsFalse", "False"),
NEGATING_SIMPLE_PROPERTY("IsNot", "Not"),
SIMPLE_PROPERTY("Is", "Equals");
```

```
List<UserDetails> findUserDetailsByNameIsIn(List<String> userName);

```

```
Hibernate:
    select
        ud1_0.user_id,
        ud1_0.user_name,
        ud1_0.phone
    from
        user_details ud1_0
    where
        ud1_0.user_name in (?)
```

Concept && Coding

```
List<UserDetails> findUserDetailsByNameLike(String userName);

```

```
Hibernate:
    select
        ud1_0.user_id,
        ud1_0.user_name,
        ud1_0.phone
    from
        user_details ud1_0
    where
        ud1_0.user_name like ? escape '\'
```

Delete:

- Need to add @Transactional annotation.

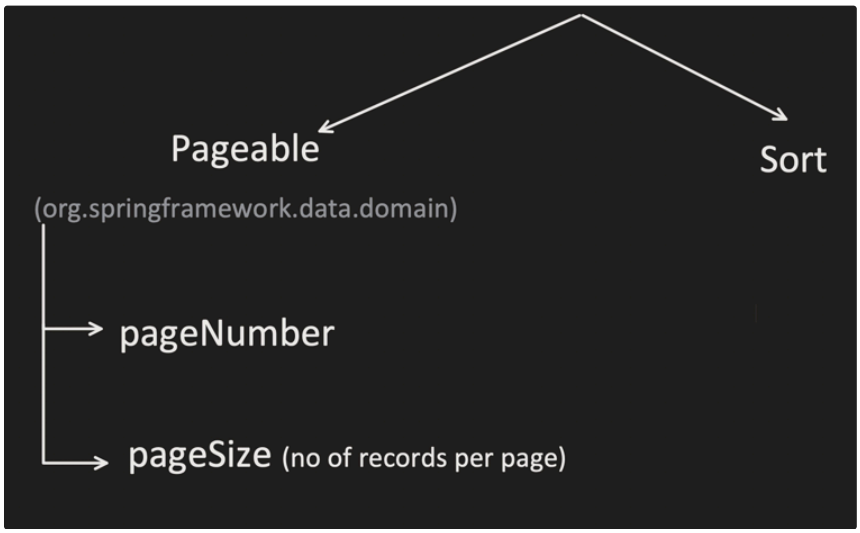
```
@Transactional
void deleteByName(String userName);

```

```
Hibernate:
    select
        ud1_0.user_id,
        ud1_0.user_name,
        ud1_0.phone
    from
        user_details ud1_0
    where
        ud1_0.user_name=?
Hibernate:
    delete
    from
        user_details
    where
        user_id=?
Hibernate:
    delete
    from
        user_details
    where
        user_id=?
Hibernate:
    delete
    from
        user_details
    where
        user_id=?
```

Concept && Coding

Paginations and Sorting in Derived Query:

- JPA provides 2 interfaces to support Pagination and Sorting i.e.




```
@Repository
```

```java
public interface UserDetailsRepository extends
        JpaRepository<UserDetails, Long> {

    List<UserDetails> findUserDetailsByNameStartingWith(String userName, Pageable page);

}

```

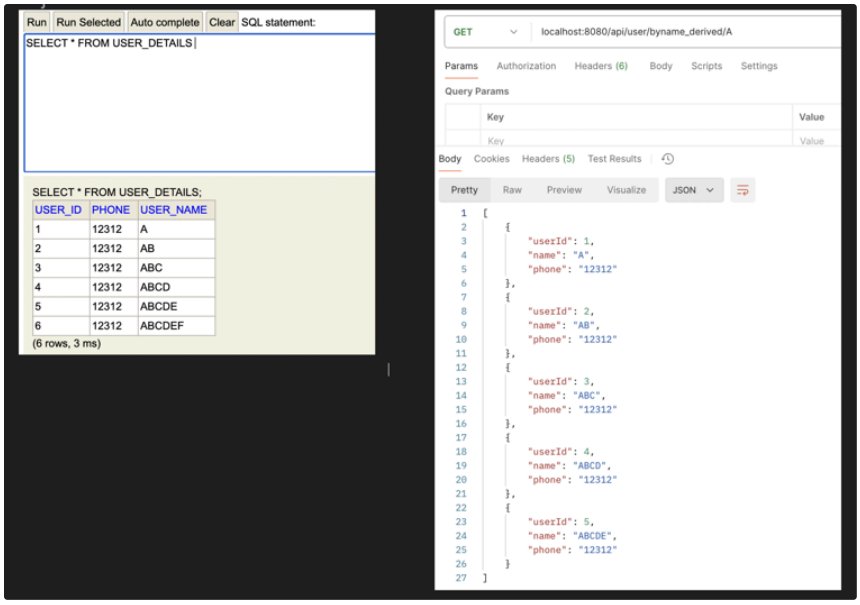
```java
@Service
public class UserDetailsService {

    @Autowired
    UserDetailsRepository userDetailsRepository;

    public UserDetails saveUser(UserDetails user) {
        return userDetailsRepository.save(user);
    }

    public List<UserDetails> findByNameDerived(String name) {
        Pageable pageable = PageRequest.of(0, 5); // Page 0, 5 records per page
        return userDetailsRepository.findUserDetailsByNameStartingWith(name, pageable);
    }
}

```

If we need more info about Pages, then we can use "Page" as return type

```java
@Repository
public interface UserDetailsRepository extends
        JpaRepository<UserDetails, Long> {

    Page<UserDetails> findUserDetailsByNameStartingWith(String userName, Pageable page);
}

```

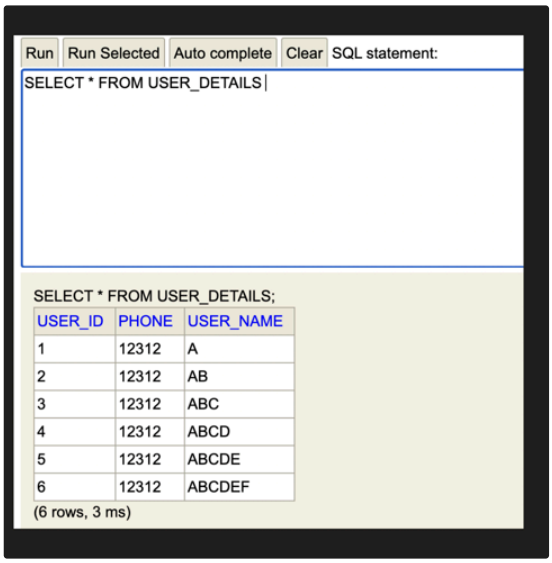
Concept && Coding

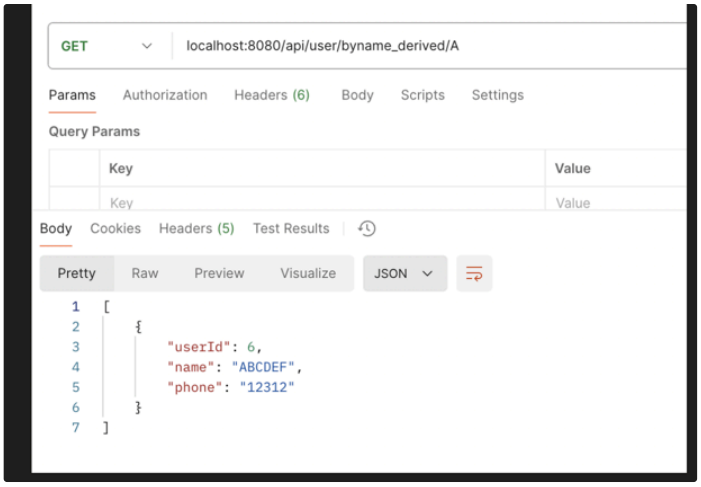
```java
public List<UserDetails> findByNameDerived(String name) {

    Pageable pageable = PageRequest.of(0, 5); // Page 0, 5 records per page

    Page<UserDetails> userDetailsPage =

    userDetailsRepository.findUserDetailsByNameStartingWith(name, pageable);

    List<UserDetails> userDetailsList = userDetailsPage.getContent();

    System.out.println("total pages: " + userDetailsPage.getTotalPages());
    System.out.println("is first page: " + userDetailsPage.isFirst());
    System.out.println("is last page: " + userDetailsPage.isLast());

    return userDetailsList;
}

```




```java
public List<UserDetails> findByNameDerived(String name) {
    Pageable pageable = PageRequest.of(1, 5); // Page 0, 5 records per page
    Page<UserDetails> userDetailsPage =
    userDetailsRepository.findUserDetailsByNameStartingWith(name, pageable);

    List<UserDetails> userDetailsList = userDetailsPage.getContent();

    System.out.println("total pages: " + userDetailsPage.getTotalPages());
    System.out.println("is first page: " + userDetailsPage.isFirst());
    System.out.println("is last page: " + userDetailsPage.isLast());

    return userDetailsList;
}
```

Concept && Coding



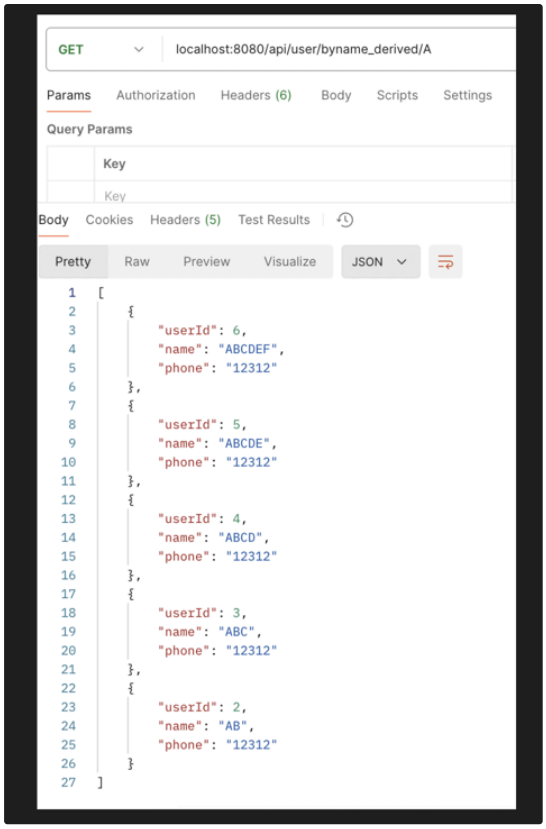



```
Hibernate:
    select
        ud1_0.user_id,
        ud1_0.user_name,
        ud1_0.phone
    from
        user_details ud1_0
    where
        ud1_0.user_name like ? escape '\'
    offset
        ? rows
    fetch
        first ? rows only
total pages: 2
is first page: false
is last page: true
```

Concept && Coding

Paginations with Sorting:

```java
public List<UserDetails> findByNameDerived(String name) {
    Pageable pageable = PageRequest.of(0, 5, Sort.by("name").descending()); // Page 0, 5 records per page
    Page<UserDetails> userDetailsPage = userDetailsRepository.findUserDetailsByNameStartingWith(name, pageable);

    List<UserDetails> userDetailsList = userDetailsPage.getContent();

    System.out.println("total pages: " + userDetailsPage.getTotalPages());
    System.out.println("is first page: " + userDetailsPage.isFirst());
    System.out.println("is last page: " + userDetailsPage.isLast());

    return userDetailsList;
}

```

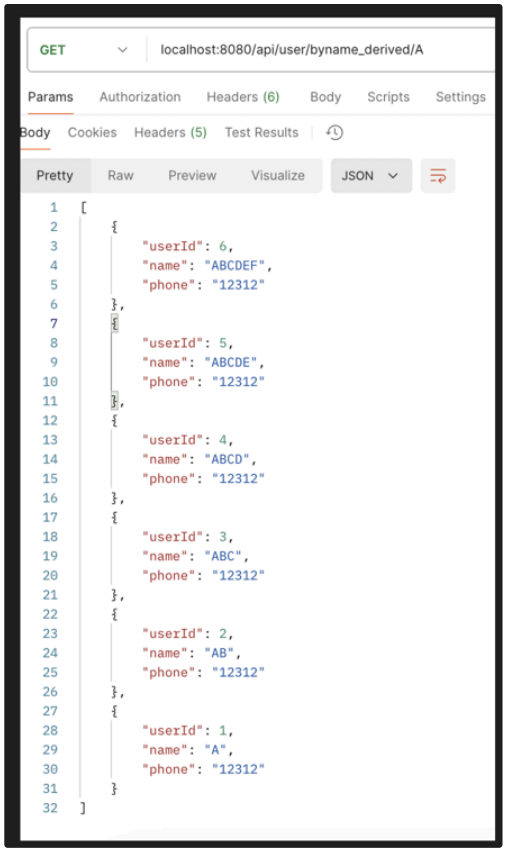


Only Sorting:

Concept && Coding

```java
@Repository
public interface UserDetailsRepository extends JpaRepository<UserDetails, Long> {

    List<UserDetails> findUserDetailsByNameStartingWith(String userName, Sort sort);

}

```

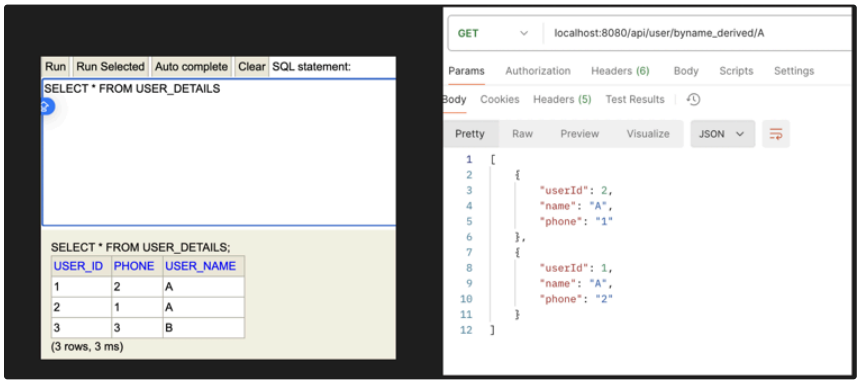
```java
public List<UserDetails> findByNameDerived(String name) {

    return userDetailsRepository.findUserDetailsByNameStartingWith(
            name,
            Sort.by("name").descending()
    );
}

```



- sort.by accepts multiple fields.
- When multiple fields provided, sorting applied in order.
- first it sort by first field and if there are duplicates then second field is used and so on.

Concept && Coding

```
public List<UserDetails> findByNameDerived(String name) {

    return userDetailsRepository.findUserDetailsByNameStartingWith(
            name,
            Sort.by("name", "phone").ascending()
    );
}

```



If we need different sorting order for different fields

```
public List<UserDetails> findByNameDerived(String name) {

    Sort sort = Sort.by(
            Sort.Order.asc("name"),
```

```
            Sort.Order.desc("phone")
    );

    return userDetailsRepository.findUserDetailsByNameStartingWith(name, sort);
}

```

Queries which are little complex and can't be handled via Derived Query, we can use:

JPQL:

- Java Persistence Query Language.
- Similar to SQL but works on ***Entity Object*** instead of direct database.
  - Its database independent
  - Works with Entity name and fields and not with table column names.

Syntax:

```
@Query("SELECT u FROM UserDetails u WHERE u.name = :userFirstName")
  /*
  u:-Entity alias, returns all the fields
  userDetails:-This is an entity, not a table name
  u.name:-This is an entity field name,
  not a column name
    List<user details>:-There is no strict rule for Return type:
    - you can return List or
    -Single object
    But, if say there are more than one rows, but in return type,
    we return Single Object, then JPQL will throw an exception
    userFirstName:-Binds, method parameter
      with named userFirstName:- parameter in the query
  */
List<UserDetails> findByUserName(@Param("userFirstName") String userName);

```

Concept && Coding

JPQL query with JOIN

OneToOne

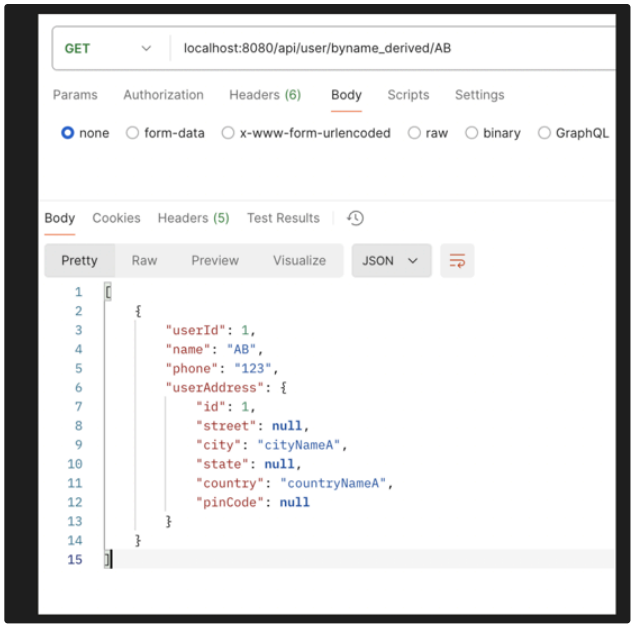
```
@Table(name = "user_details")
@Entity
public class UserDetails {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long userId;

    @Column(name = "user_name")
    private String name;
    private String phone;

    @OneToOne(cascade = CascadeType.ALL)
    @JoinColumn(name = "user_address")
    private UserAddress userAddress;

    //getters and setters
}

```

```java
@Entity
@Table(name = "user_address")
public class UserAddress {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private String street;
    private String city;
    private String state;
    private String country;
    private String pinCode;

    //getters and setters
}

```

```java
@Repository
public interface UserDetailsRepository extends
        JpaRepository<UserDetails, Long> {
  /*
      We don't specifically need to put "On" here,
      JPA will automatically do that
  */

    @Query("SELECT ud FROM UserDetails ud JOIN ud.userAddress ad WHERE ud.name = :userFirstName")
    List<UserDetails> findUserDetailsWithAddress(
            @Param("userFirstName") String userName
    );
}

```

Concept && Coding

```
Hibernate:
    select
        ud1_0.user_id,
        ud1_0.user_name,
        ud1_0.phone,
        ud1_0.user_address
    from
        user_details ud1_0
    join
        user_address ua1_0
          on ua1_0.id=ud1_0.user_address
    where
        ud1_0.user_name=?
```




```java
@Repository
public interface UserDetailsRepository extends
        JpaRepository<UserDetails, Long> {

    @Query("SELECT ud.name, ad.country FROM UserDetails ud JOIN ud.userAddress ad WHERE ud.name = :userFirstName")
    List<Object[]> findUserDetailsWithAddress(@Param("userFirstName") String userName);

}

```

Concept && Coding

```java
public class UserDTO {

    String userName;
    String country;

    // Constructor to populate from UserDetails entity
    public UserDTO(String userName, String country) {
        this.userName = userName;
        this.country = country;
    }

    //getters and setters
}

```

```java
public List<UserDTO> findByNameDerived(String name) {

    List<Object[]> dbOutput = userDetailsRepository.findUserDetailsWithAddress(name);
    List<UserDTO> output = new ArrayList<>();

    for (Object[] val : dbOutput) {
        String userName = (String) val[0];
        String country = (String) val[1];
        UserDTO dto = new UserDTO(userName, country);
        output.add(dto);
    }

    return output;
}

```

If we don't, want Object[] to be used, we can also return direct custom DTO

```
@Repository
public interface UserDetailsRepository extends
        JpaRepository<UserDetails, Long> {

    @Query("SELECT new com.conceptandcoding.learningspringboot.jpa.DTO.UserDTO(ud.name, ad.country) " +
            "FROM UserDetails ud JOIN ud.userAddress ad WHERE ud.name = :userFirstName")
    List<UserDTO> findUserDetailsWithAddress(@Param("userFirstName") String userName);

}

```

- **OneToMany**

```
@Table(name = "user_details")
@Entity
public class UserDetails {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long userId;

    @Column(name = "user_name")
    private String name;
    private String phone;

    @OneToMany(cascade = CascadeType.ALL)
    @JoinColumn(name = "user_id") //fk in user address table
    private List<UserAddress> userAddressList = new ArrayList<>();

    //getters and setters
}

```

Concept && Coding

```
@Entity
@Table(name = "user_address")
public class UserAddress {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private String street;
    private String city;
    private String state;
    private String country;
    private String pinCode;

    //getters and setters
}

```

```
@Query("SELECT ud FROM UserDetails ud JOIN ud.userAddressList ad WHERE ud.name = :userFirstName")
List<UserDetails> findUserDetailsWithAddress(@Param("userFirstName") String userName);

```

N+1 Problem and its Solution:

Problem :

Say, 1 User can have Many Addresses.
And our Query is such that, it can fetch more than 1 Users. Then this problem can occurs.

So, say we have 'N' Users. Then below queries will be hit by JPA:

- 1 query to fetch all the USERS.
- For each User it will fetch ADDRESSES, so for N users, it will fetch N times.

So total number of query hit : N+1.

So we need to find the way, so that only 1 QUERY it hit instead of N+1.

Before going for the solution for this problem, One question might be coming to our mind:

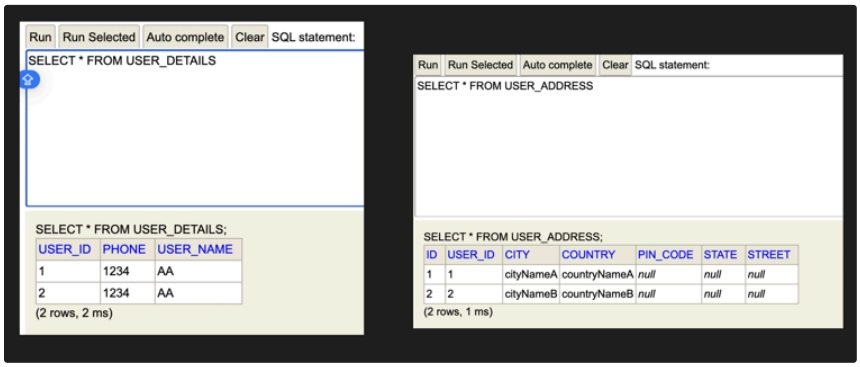
What if, we use EAGER initialization, then can we avoid this issue?

Concept && Coding

NO because EAGER initialization do not work, when our query tries to fetch multiple PARENT rows and that also have multiple CHILD.

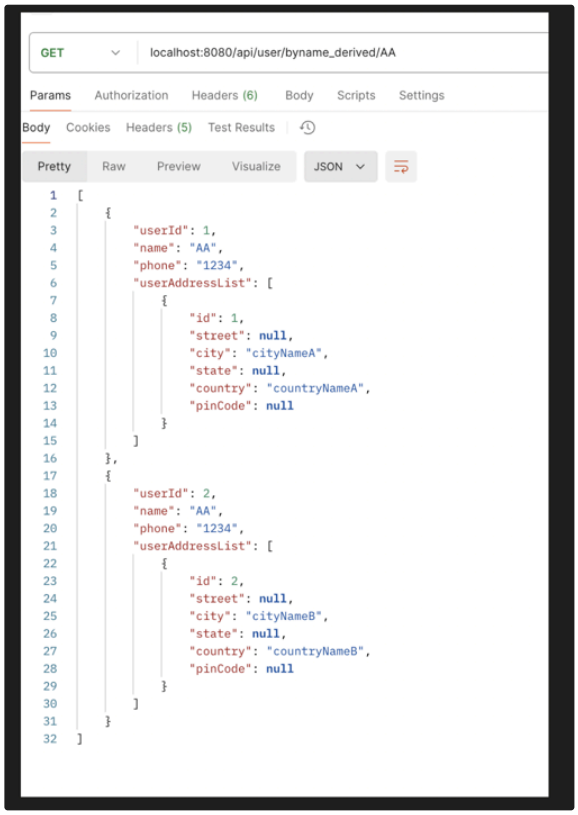
In previous video, we tested EAGER with ***"findByID(id)"*** method, in which it make sure that, our query is fetching only 1 PARENT and that can have many CHILD, that's fine. In that JPA internally draft a JOIN query.

But when Multiple parent with Multiple child get involved, EAGER do not work in just 1 query, it first fetches all the parent and then for each parent, it fetch all its child.




```
@Query("SELECT ud FROM UserDetails ud JOIN ud.userAddressList ad WHERE ud.name = :userFirstName")
```

```
List<UserDetails> findUserDetailsWithAddress(@Param("userFirstName")
String userName);
```



Concept && Coding

So, how to solve this, N+1 problem?

Solution1: using ***JOIN FETCH*** (JPQL)

```java
@Query("SELECT ud FROM UserDetails ud JOIN FETCH ud.userAddressList ad WHERE ud.name = :userFirstName")
List<UserDetails> findUserDetailsWithAddress(
        @Param("userFirstName") String userName
);

```

```
Hibernate:
    select
        ud1_0.user_id,
        ud1_0.user_name,
        ud1_0.phone,
        ual1_0.user_id,
        ual1_0.id,
        ual1_0.city,
        ual1_0.country,
        ual1_0.pin_code,
        ual1_0.state,
        ual1_0.street
    from
        user_details ud1_0
    join
        user_address ual1_0
            on ud1_0.user_id=ual1_0.user_id
    where
        ud1_0.user_name=?
```

GET localhost:8080/api/user/byname_derived/AA

Concept && Coding

Params Authorization Headers (6) Body Scripts Settings

Body Cookies Headers (5) Test Results

Pretty Raw Preview Visualize JSON


```
[
    {
        "userId": 1,
        "name": "AA",
        "phone": "1234",
        "userAddressList": [
            {
                "id": 1,
                "street": null,
                "city": "cityNameB",
                "state": null,
                "country": "countryNameB",
                "pinCode": null
            }
        ]
    },
    {
        "userId": 2,
        "name": "AA",
        "phone": "1234",
        "userAddressList": [
            {
                "id": 2,
                "street": null,
                "city": "cityNameA",
                "state": null,
                "country": "countryNameA",
                "pinCode": null
            }
        ]
    }
]
```


Solution2: using ***@BatchSize(size=10)***

- It wont make only 1 query, but it will reduce it, as it will divide it into batches

```java
@Table(name = "user_details")
@Entity
public class UserDetails {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long userId;

    @Column(name = "user_name")
    private String name;
    private String phone;

    @OneToMany(cascade = CascadeType.ALL, fetch = FetchType.EAGER)
    @BatchSize(size = 10)
    @JoinColumn(name = "user_id") //fk in user address table
    private List<UserAddress> userAddressList;

    //getters and setters
}

```

```
Hibernate:
    select
        ud1_0.user_id,
        ud1_0.user_name,
        ud1_0.phone
    from
        user_details ud1_0
    join
        user_address ual1_0
            on ud1_0.user_id=ual1_0.user_id
    where
        ud1_0.user_name=?
Hibernate:
    select
        ual1_0.user_id,
        ual1_0.id,
        ual1_0.city,
        ual1_0.country,
        ual1_0.pin_code,
        ual1_0.state,
        ual1_0.street
    from
        user_address ual1_0
    where
        ual1_0.user_id in (?, ?, ?, ?, ?, ?, ?, ?, ?, ?)
```

Concept && Coding

Solution3: using ***@EntityGraph(attributePaths="userAddressList")***

- Used over method (helpful in derived methods)
- Tell JPA to fetch all the entries of UserAddress along with user details.

```java
@EntityGraph(attributePaths = "userAddressList")
List<UserDetails> findUsersBy();

```

How to join Many tables?

Its almost same as SQL only

Say, we have
Table A has one to many relationship with Table B

Table B has one to many relationship with Table C

@Query("SELECT a FROM A a JOIN a.bList b JOIN b.cList c WHERE c.someProperty = :someValue")
List<A> findAWithBAndC(@Param("someValue") String someValue);

@Modifying Annotation

- when @Query annotation used, by-default JPA expects ***SELECT*** query.
- If we try to use "DELETE" or "INSERT" or "UPDATE" query with @Query, JPA will throw error, that:

```
query.IllegalSelectQueryException Create breakpoint : Expecting a SELECT Query [org.hibernate.query.sqm.tree.select.SqmSelectStatement],
ernate.query.sqm.internal.SqmUtil.verifyIsSelectStatement(SqmUtil.java:109) ~[hibernate-core-6.5.2.Final.jar:6.5.2.Final]
ernate.query.sqm.internal.QuerySqmImpl.verifySelect(QuerySqmImpl.java:494) ~[hibernate-core-6.5.2.Final.jar:6.5.2.Final]
```

- @Modifying annotation, is to tell JPA that, expect either "DELETE" or "INSERT" or "UPDATE" query with @Query

- Since we are trying to update the DB, we also need to use @Transactional annotation.

```
@Modifying
@Transactional
@Query("DELETE FROM UserDetails ud WHERE ud.name = :userFirstName")
void deleteByUserName(@Param("userFirstName") String userName);

```

Concept && Coding

Understanding Usage of Flush and Clear:

- As we know, Flush just pushed the persistence context changes to DB but hold the value in persistence context.

- Clear, purge the persistence context, and required fresh DB call

```
@Modifying
@Query("DELETE FROM UserDetails ud WHERE ud.name = :userFirstName")
void deleteByUserName(@Param("userFirstName") String userName);

```

```
@Service
public class UserDetailsService {

    @Autowired
    UserDetailsRepository userDetailsRepository;

    public UserDetails saveUser(UserDetails user) {
        return userDetailsRepository.save(user);
    }

    @Transactional
```

```java
    public void deleteByUserName(String name) {
        userDetailsRepository.findById(1L).get();
        userDetailsRepository.deleteByUserName(name);
        Optional<UserDetails> output =
userDetailsRepository.findById(1L);
        System.out.println("output present: " + output.isPresent());
    }
}

```

```
Hibernate:
    select
        ud1_0.user_id,
        ud1_0.user_name,
        ud1_0.phone,
        ua1_0.id,
        ua1_0.city,
        ua1_0.country,
        ua1_0.pin_code,
        ua1_0.state,
        ua1_0.street
    from
        user_details ud1_0
    left join
        user_address ua1_0
            on ua1_0.id=ud1_0.user_address_id
    where
        ud1_0.user_id=?
Hibernate:
    delete
    from
        user_details ud1_0
    where
        ud1_0.user_name=?
output present: true
```

Concept && Coding

Now using, Flush and Clear

```java
@Modifying(flushAutomatically = true, clearAutomatically = true)
@Query("DELETE FROM UserDetails ud WHERE ud.name = :userFirstName")
void deleteByUserName(@Param("userFirstName") String userName);

```

```java
@Service
public class UserDetailsService {

    @Autowired
    UserDetailsRepository userDetailsRepository;

    public UserDetails saveUser(UserDetails user) {
        return userDetailsRepository.save(user);
    }

    @Transactional
    public void deleteByUserName(String name) {
        userDetailsRepository.findById(1L).get();
        userDetailsRepository.deleteByUserName(name);
        Optional<UserDetails> output =
userDetailsRepository.findById(1L);
        System.out.println("output present: " + output.isPresent());
```

```
    }

}

```

```
Hibernate:
    select
        ud1_0.user_id,
        ud1_0.user_name,
        ud1_0.phone,
        ua1_0.id,
        ua1_0.city,
        ua1_0.country,
        ua1_0.pin_code,
        ua1_0.state,
        ua1_0.street
    from
        user_details ud1_0
    left join
        user_address ua1_0
            on ua1_0.id=ud1_0.user_address_id
    where
        ud1_0.user_id=?
Hibernate:
    delete
    from
        user_details ud1_0
    where
        ud1_0.user_name=?
Hibernate:
    select
        ud1_0.user_id,
        ud1_0.user_name,
        ud1_0.phone,
        ua1_0.id,
        ua1_0.city,
        ua1_0.country,
        ua1_0.pin_code,
        ua1_0.state,
        ua1_0.street
    from
        user_details ud1_0
    left join
        user_address ua1_0
            on ua1_0.id=ud1_0.user_address_id
    where
        ud1_0.user_id=?
output present: false
```

Concept && Coding

Pagination and Sorting in JPQL

Same like discussed in derived query method

```
@Query("SELECT ud FROM UserDetails ud WHERE ud.name = :userFirstName")
List<UserDetails> findUserDetails(@Param("userFirstName") String
userName, Pageable pageable);

```

```
public List<UserDetails> findByUserName(String name) {
    Pageable page = PageRequest.of(pageNumber: 1, pageSize: 5);
    return userDetailsRepository.findUserDetails(name, page);
}

```

```
Hibernate:
    select
        ud1_0.user_id,
        ud1_0.user_name,
        ud1_0.phone,
        ud1_0.user_address_id
    from
        user_details ud1_0
    where
        ud1_0.user_name=?
    offset
        ? rows
    fetch
        first ? rows only
```

@NamedQuery Annotation

- We can name our Query, so that we can reuse it.

```java
@Table(name = "user_details")
@Entity
@NamedQuery(
    name = "findByUserName",
    query = "SELECT u FROM UserDetails u WHERE u.name = :userFirstName"
)
public class UserDetails {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long userId;

    @Column(name = "user_name")
    private String name;
    private String phone;

    @OneToOne(cascade = CascadeType.ALL)
    private UserAddress userAddress;

    //getters and setters
}

```

Concept && Coding

```java
@Query(name = "findByUserName")
List<UserDetails> findUserDetails(@Param("userFirstName") String userName, Pageable pageable);
```